Forests obtained by contracting an involution

Human Compression project: a component of the L044 recovery

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We isolate an elementary reduction for the two-block case in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 3.5. Contraction below produces a graph. It does not require every generator to permute the contracted pairs. The final equality exclusions are separate obligations.

Hypotheses and notation. Let $I = \{0, \dots, r-1\} \setminus \{i\}$, and let $H = \langle \rho_j : j \in I \rangle$ be transitive on two preserved blocks B_0, B_1, each of size m. Every ρ_j is an even involution, and $\rho_j \rho_k = \rho_k \rho_j$ for $|j - k| > 1$. Suppose $H_j = \langle \rho_k : k \in I \setminus \{j\} \rangle$ is intransitive for every $j \in I$. Choose $\ell \in I$ such that $l = \rho_\ell$ interchanges the blocks. Its orbits are m pairs, each meeting both blocks once. In particular m is even. Put

$$C = \{j \in I \setminus \{\ell\} : \rho_j l = l \rho_j\}, \qquad R = I \setminus (C \cup \{\ell\}).$$

The string relations imply $|R| \leq 2$. For an involution, an actual edge means one of its transpositions. Contract an actual edge to the unordered pair of its l-orbits, deleting loops. Multiplicity is retained only when explicitly stated. An R generator need not induce a permutation of these m vertices.

For each $j \in I \setminus \{\ell\}$ choose an actual j-edge joining different H_j-orbits; it exists by transitivity of H. Every H_j-orbit is a union of l-pairs. Let e_j be the contracted edge so chosen, and let F consist of these $r - 2$ edges on all m vertices.

Lemma 1. *The graph F is a forest, with no parallel edges. If an actual h-edge, $h \in I \setminus \{\ell\}$, joins two different vertices in one component of F, those vertices are precisely the endpoints of e_h.*

Proof. Every walk in the contracted graph using labels other than j lifts to an actual walk using those labels and l: within each intermediate pair insert an l-edge if necessary to reach the endpoint of the next actual edge. The endpoints can likewise be adjusted to any prescribed points in their pairs. Since $l \in H_j$, such a walk cannot join the endpoints of the chosen actual j-edge. Any cycle of selected edges would give precisely such a walk by removing its unique j-edge. This also excludes a cycle of two parallel selected edges.

Now take the unique F-path between the endpoints of a contracted h-edge. If that path contained e_j with $j \neq h$, the new h-edge and the rest of the path would bypass e_j without using j, a contradiction by the same lifting argument. The path is therefore the single edge e_h, since only one edge of that label was selected. □

Proposition 2. *If $R \neq \varnothing$, then $r \leq m$. At equality F has exactly two components.*

Proof. A generator $h = \rho_j$ with $j \in R$ joins at least two distinct unordered pairs of l-pairs. To see this, first suppose h preserves each block. There are no actual h-edges inside an l-pair. If there were just one nonloop contracted edge, its multiplicity would be one or two. Evenness forces two, and the resulting simultaneous transpositions on the two rows commute with l.

If h interchanges the blocks, it is a perfect matching. A single nonloop contracted edge must be doubled: once one point of a pair is matched outside that pair, its other point cannot be

matched internally. On all other pairs h equals l. This again commutes with l. Zero nonloop edges also imply commutation in both cases.

If F were connected, Lemma 1 would allow only the single contracted edge e_j for this generator, contrary to the preceding paragraph. Thus F has at least two components, whence $r - 2 \leq m - 2$. Equality gives exactly two components. □

Lemma 3. *Suppose $r = m$ and $R \neq \varnothing$. For any $h \in I \setminus \{\ell\}$, its simple contracted graph consists of e_h alone, e_h and one edge across the two components of F, or a triangle containing e_h and one vertex in the other component. For $h \in R$ there are precisely the following forms.*

1. *If ρ_h preserves the blocks, it is a double transposition. Its two contracted edges are distinct, one selected and one crossing the component cut.*

2. *If ρ_h interchanges the blocks, its contracted graph is a triangle with multiplicity one on each edge. On every other l-pair it equals l. On the three exceptional pairs, $l\rho_h$ acts by a three-cycle on one row and its inverse on the other.*

Proof. Write $e_h = ab$ in one component of F. All additional distinct contracted h-edges cross the component cut by Lemma 1. Given two such edges cd, ef, orient them with c, e in the first tree and d, f in the second. The tree paths from c to e and from d to f, together with these two edges, form a cycle. Either tree path is allowed to have length zero. Every selected edge on these paths must have label h, by the lifting argument. The path in the component not containing ab therefore has length zero, and the other path consists of ab. The two new edges are consequently ax, bx for one vertex x in the other tree. A third distinct crossing edge cannot satisfy this condition with each of those two edges. This proves the first assertion.

For a block-preserving generator, the contracted edges are the union of the matchings in the two rows. When the simple edges are distinct this gives a proper colouring by two colours, so it cannot contain an odd cycle. A generator in R thus has exactly two simple edges. Their multiplicities are at most two. Evenness allows $(1, 1)$ or $(2, 2)$; the latter interchanges whole pairs and commutes with l. Only $(1, 1)$ remains. There can be no other transpositions inside pairs, since these would interchange rows.

For a block-interchanging generator, each pair has either an internal matching edge or two outgoing incidences. Its nonloop contracted multigraph is therefore 2-regular on its nonisolated vertices. With at most two simple edges it would consist of doubled disjoint edges, giving commutation with l. Hence it is the triangle, with every multiplicity one. Outside it the matching is internal. Choose coordinates $(v, 0), (v, 1)$ with l the row interchange. An involution interchanging rows has the form $(v, 0) \mapsto (p(v), 1)$, $(v, 1) \mapsto (p^{-1}(v), 0)$. Here p is a three-cycle on the exceptional vertices and fixes all others. The asserted description of $l\rho_h$ follows. □

Proposition 4. *At the equality in Lemma 3, let J contain all simple contracted edges of the R generators, on all m vertices, including isolates. Let $E_C = \{e_j : j \in C\}$. Every edge of E_C is a bridge of $J \cup E_C$. If $|R| = 2$, then $\operatorname{rank}(J) = 3$ and $J \cup E_C$ is connected. Here graph rank is the number of vertices minus the number of components. If $|R| = 1$, then $J \cup E_C$ is also connected; it is a tree when the sole R generator preserves the two blocks.*

Proof. Any alternate path around e_j, $j \in C$, in $J \cup E_C$ uses only R labels and the other selected C labels. It lifts to a forbidden H_j walk. The same argument excludes a duplicate of e_j among the edges of J. Adding the $|C| = m - 2 - |R|$ bridges thus raises graph rank by exactly $|C|$. Since the final rank is at most $m - 1$, we have $\operatorname{rank}(J) \leq |R| + 1$.

Each individual R graph has rank two by Lemma 3. For $|R| = 1$, adding the $m - 3$ bridges gives rank $m - 1$, hence connectedness. In the block-preserving case J has just two edges and is a forest; adjoining bridges gives a tree.

For $|R| = 2$, suppose their union still had rank two. The component partition of each individual graph must then equal that of the union. Either both graphs consist of the same two

disjoint edges, or both are supported on the same three vertices. In the first case exactly one edge is internal to an F-component, so both labels must select that same edge. This contradicts Lemma 1. In the second case the three vertices cannot all lie in one F-component, because each R graph has a crossing edge. Their distribution is $2 + 1$, with exactly one internal unordered pair. Both graphs must contain and select that pair, giving the same contradiction. Thus $\operatorname{rank}(J) > 2$. The upper bound already proved gives rank three; adjoining the $m - 4$ bridges gives rank $m - 1$ and connectedness. □

Attribution and scope. The chosen edges between omission orbits and the use of string relations belong to the source argument. The present proof supplies the explicit walk lifting, local support statements and bridge-rank reduction, without assuming a quotient action on the pairs. These steps were written by the root agent and checked in working form by a distinct agent. No computational census is a premise. This component deliberately assumes $R \neq \varnothing$ for its rank bound. The separate empty-R reduction requires the parent alternating C-group, the acquired transitive rank theorem, finite bases and the smaller-degree induction in the pair-block theorem; preservation of pairs alone does not justify that invocation. No whole-original recovery, open-conjecture solution, formal proof certificate or numerical compression claim follows here.